Bean-Leaf Disease Classification with SigLIP 2 and DINOv2: Accuracy, Adaptation and Computational Cost

Experiments executed: 24 October 2026.

1. Introduction and research question

This study compares the recent SigLIP 2 (2025) vision-language encoder with DINOv2 (2023) for bean-leaf classification [1,2]. Both use attention over image-patch representations learned through pretraining.

The primary question was whether partial SigLIP 2 fine-tuning improved macro-F1 over partial DINOv2. Frozen features, adaptation benefits, cost and failures provided secondary comparisons. This was an exploratory extension: original DINOv2 test results were already known. Both backbones were trained afresh under fixed settings, but the reused test set cannot provide fresh confirmatory evidence.

2. Dataset selection and audit

Makerere AI Lab's Beans dataset contains smartphone photographs from Ugandan fields, collected with the National Crops Resources Research Institute and labelled by agricultural experts as angular leaf spot, bean rust or healthy [3]. Its limited sample size motivates pretrained representations; field backgrounds provide a relevant challenge beyond isolated leaf cutouts.

The pinned Hugging Face release contains 1,295 images, although the original repository describes 1,296 [3,4]. Table 1 gives the actual loaded counts. Every image was 500 × 500 RGB, with no corrupt files. File and decoded-pixel hashes found no exact duplicates. A 64-bit perceptual-hash screen at Hamming distance <=12 produced 14 candidate pairs, including seven across splits. Recorded AI-assisted visual screening judged them distinct, so none were excluded. This was not independent expert verification of labels.

Class	Training	Validation	Test
Angular leaf spot	345	44	43
Bean rust	348	45	43
Healthy	341	44	42
Total	1,034	133	128

Table 1. Retained counts from the audited dataset release.

The published split was preserved. Test images had also been inspected during the original integrity audit. No plant or farm identifiers were available, so independence at those levels is unverified. Hash screening cannot exclude every related photograph, and overlap with either backbone's pretraining collection was not measured. Revisions, the manifest and screening decisions accompany the code.

3. Model selection and configuration

Google's public `google/siglip2-base-patch16-224` checkpoint was downloaded without authentication and its published weight hash verified [5]. The 2025 family combines image-text learning, captioning, self-distillation, masked prediction and data curation [1]. Its paper reports 78.2% ImageNet zero-shot accuracy versus original SigLIP's 76.2%, leading the same-size B/16-224 baselines in Table 1 [1]. This supports selecting a recent advanced model; it establishes neither the best October 2026 model nor state-of-the-art Beans performance.

Only SigLIP 2's image tower was used. Its 12 blocks process 196 patches of 16 × 16 pixels at 224 × 224, with 768-dimensional embeddings and 12 attention heads. Final normalisation and pretrained attention pooling produce a vector for a new three-class linear classifier [5,6]. Token zero is an image patch, not a class token. This fixed-resolution checkpoint uses `SiglipVisionModel`, distinct from NaFlex.

DINOv2 used `facebook/dinov2-small`: 12 blocks, 384-dimensional embeddings, six heads and 256 patches of 14 × 14 pixels, plus a class token. Its final normalised class token fed the classifier [2,7]. DINOv2's LVD-142M pretraining differs from SigLIP 2's WebLI pretraining. Capacity, pretraining, pooling and preprocessing differ, so this system comparison cannot isolate publication year or architecture.

Model	Total parameters	Frozen-arm trainable	Partial-arm trainable
DINOv2-S/14	22,057,731	1,155	3,552,387
SigLIP 2-B/16	92,886,531	2,307	14,179,587

Table 2. Actual loaded image-model counts, including pooling and the new classifier; the SigLIP text tower is excluded.

4. Pre-processing and training

RGB images were resized in full to 224 × 224, preserving their square aspect ratio. SigLIP 2 followed its native bilinear resize and channel mean/standard deviation 0.5 [5]. DINOv2 used bicubic interpolation, means (0.485, 0.456, 0.406) and standard deviations (0.229, 0.224, 0.225). Its official resize-to-256/centre-crop processor was deliberately replaced with full-image resizing to retain peripheral regions [7]. Both approaches reduce fine detail.

Training used horizontal and vertical flips with probability 0.5 each, then rotation uniformly within ±15°, with model-specific mean-colour fill. This varied orientation without deliberately changing lesion colour or cropping symptoms. Seed-, epoch- and image-dependent randomness matched sample order and augmentation decisions. Validation/test preprocessing was deterministic. Frozen representations were computed from augmented training images.

The frozen arm trained only the classifier. Partial adaptation additionally trained the final two blocks and final normalisation. SigLIP's pretrained pooling head remained frozen while gradients passed through it. Backbones stayed in evaluation mode with gradients enabled for intended parameters. Classifier initialisation was paired within each backbone; representation dimensions prevent identical heads across backbones.

Cross-entropy used AdamW, classifier learning rate 0.001, backbone rate 0.00001, weight decay 0.01, batch size 8, gradient clipping at norm 1 and CUDA FP16. Rates remained constant. Training allowed 10 epochs with patience 3 on strict validation macro-F1 improvement; checkpoint ties used lower loss, then earlier epoch. Three seeds (17, 42, 2026), two arms and two models completed 12 fresh runs. No hyperparameter search was performed; settings were not separately optimised for each backbone.

Full-epoch pilots passed for both models/arms; 35 implementation, protocol and analysis tests passed. Execution used an RTX 3050 Laptop GPU (4 GB), PyTorch 2.6.0/CUDA 12.4 and Transformers 4.57.1. Configurations, histories, hashes and predictions support reproduction. Controlled seeds do not guarantee bitwise CUDA reproducibility. Codex assisted implementation, analysis and drafting.

A subsequent unchanged notebook completed all 10 code cells and 12 fresh runs locally without errors; six bootstrap checks also passed. Its separate partial-model means were 97.67% (DINOv2) and 96.90% (SigLIP 2), illustrating rerun variability. These verification outputs do not replace the original comparison below. Hosted Colab remains untested.

5. Evaluation design

Macro-F1 was primary because it weights each class equally. Accuracy, class precision/recall/F1, confusion matrices and examples complemented it. The training-majority classifier supplied a baseline. All 12 runs finished before selected checkpoints were locked and new test inference began; prior test exposure still applies.

Means and sample standard deviations over three seeds describe training variability. The primary reference comparison was seed-17 partial SigLIP 2 minus partial DINOv2. A paired class-stratified bootstrap used 2,000 resamples, seed 20261023 and shared image indices for a descriptive 95% percentile interval. This concerns that pair, not the three-seed mean; unknown plant/farm dependence remains uncorrected.

Raw training totals/means and peak allocated CUDA memory quantify measured cost. SigLIP's frozen seed-17 epoch 9 recorded 809.10 seconds while its other epochs took about 13 seconds; the cause is unknown. Before new test evaluation, a documented descriptive amendment added observed epoch medians, quartiles and ranges. Figure 1 uses pooled epoch medians and interquartile ranges; all raw durations remain retained, with no exclusion or correction. These are observed durations, not controlled hardware benchmarks.

Selected checkpoints were scored on clean training/validation images, reusing saved test scores. Online augmented training loss is not directly comparable to clean validation loss. Reference-seed blur (radius 1) and brightness ×0.7 examined synthetic sensitivity, not field robustness.

6. Results and interpretation

Both partial models averaged 96.90% macro-F1, differing by less than 0.01 percentage points (Table 3). SigLIP's frozen features exceeded frozen DINOv2 by 6.18 points; partial adaptation improved DINOv2 by 6.18 points but produced no meaningful average gain for SigLIP.

Configuration	Accuracy (%)	Macro-F1 (%)	Mean training (s)
DINOv2 frozen	90.62 ± 2.07	90.71 ± 2.05	102.8
DINOv2 partial	96.88 ± 1.56	96.90 ± 1.55	96.2
SigLIP 2 frozen	96.88 ± 1.35	96.90 ± 1.34	387.5
SigLIP 2 partial	96.88 ± 0.78	96.90 ± 0.78	195.9

Table 3. Means ± sample SD across three seeds. Raw training times exclude validation/loading/checkpoint overhead and retain long observations. The majority baseline scored 33.59% accuracy and 16.76% macro-F1.

Partial SigLIP-minus-DINO differences were -0.78, +1.55 and -0.77 points for seeds 17, 42 and 2026. The reference bootstrap interval was [-3.13, +1.57] points, including zero. Neither the seed pattern nor this interval establishes superiority.

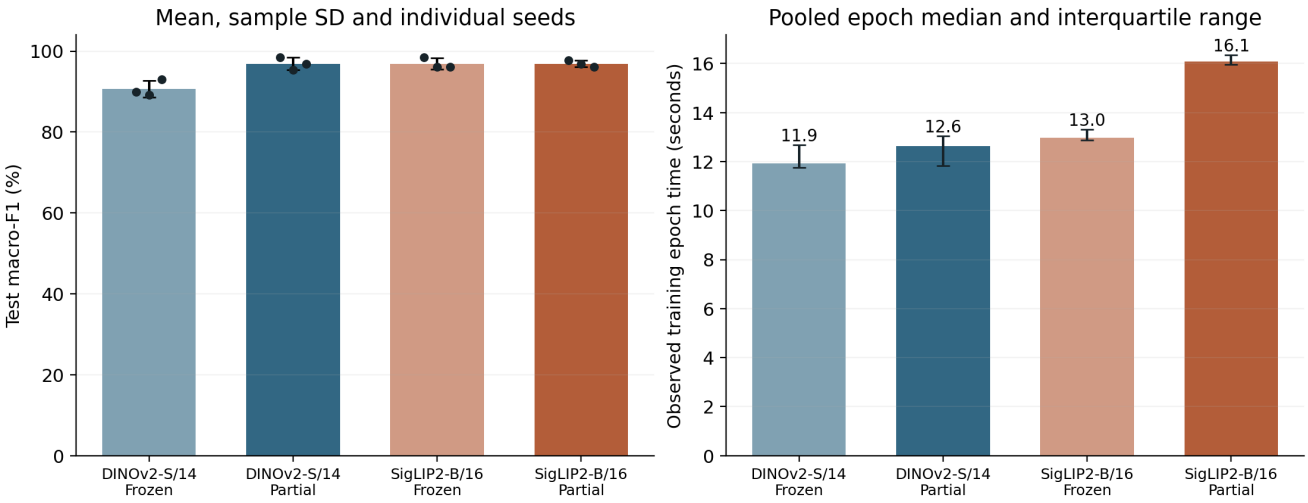


Figure 1. Test macro-F1 means, seed standard deviations and individual runs; median observed training epoch time with interquartile range. Raw totals and ranges remain in the accompanying results.

Partial DINOv2/SigLIP median epoch times were 12.62/16.08 seconds, with peak allocated memory 288/722 MB. SigLIP's extra capacity did not yield higher mean accuracy. Long durations included 809.10 seconds for frozen SigLIP and 184.36 seconds for partial SigLIP; causes remain unknown. Raw totals should not be treated as controlled speed comparisons.

Clean train/validation/test macro-F1 averaged 98.17/98.51/96.90% for partial DINOv2 and 98.91/98.26/96.90% for partial SigLIP: training-test gaps of 1.27 and 2.01 points. Later epochs showed deterioration; SigLIP seed 17 validation loss rose from 0.062 at selected epoch 6 to 0.375 at epoch 9 while training loss fell, consistent with overfitting or instability. Validation selection retained the earlier checkpoint. Small selected-checkpoint gaps cannot exclude overfitting; validation informed selection. Supplementary curves show every run.

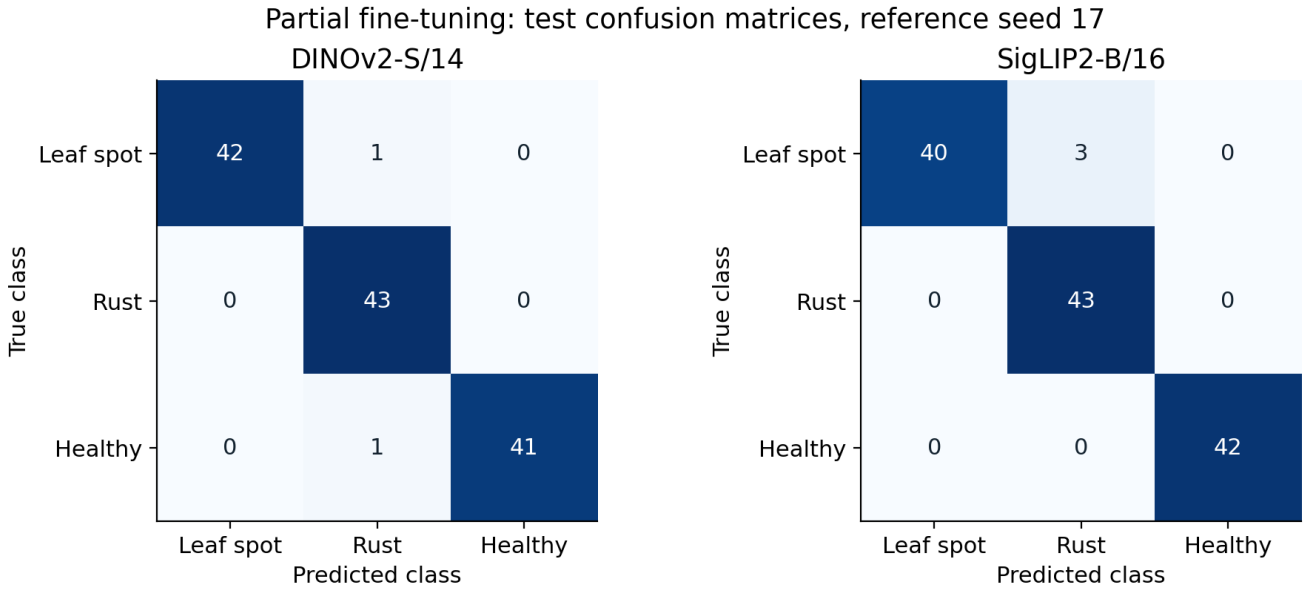


Figure 2. Partial-model test confusion matrices for reference seed 17; entries are image counts.

Reference SigLIP misclassified three leaf-spot images as rust; class F1 was 96.39%, 96.63% and 100%. Rust precision was 93.48%, recall 100%; leaf-spot recall was 93.02%. DINOv2 made two errors, with class F1 98.82%, 97.73% and 98.80%. Neither labelled a diseased leaf healthy (0/86); healthy-as-diseased counts were 0/42 for SigLIP and 1/42 for DINOv2. One SigLIP error had 98.65% confidence. All per-class metrics accompany the code.

Under blur/darkening, reference SigLIP macro-F1 was 96.07%/97.67%, versus 97.67% clean; DINOv2 scored 97.67%/98.45%, versus 98.45%. These limited checks cannot establish field reliability. With 128 test images, each error changes accuracy by 0.78 points; three seeds and no external validation limit conclusions.

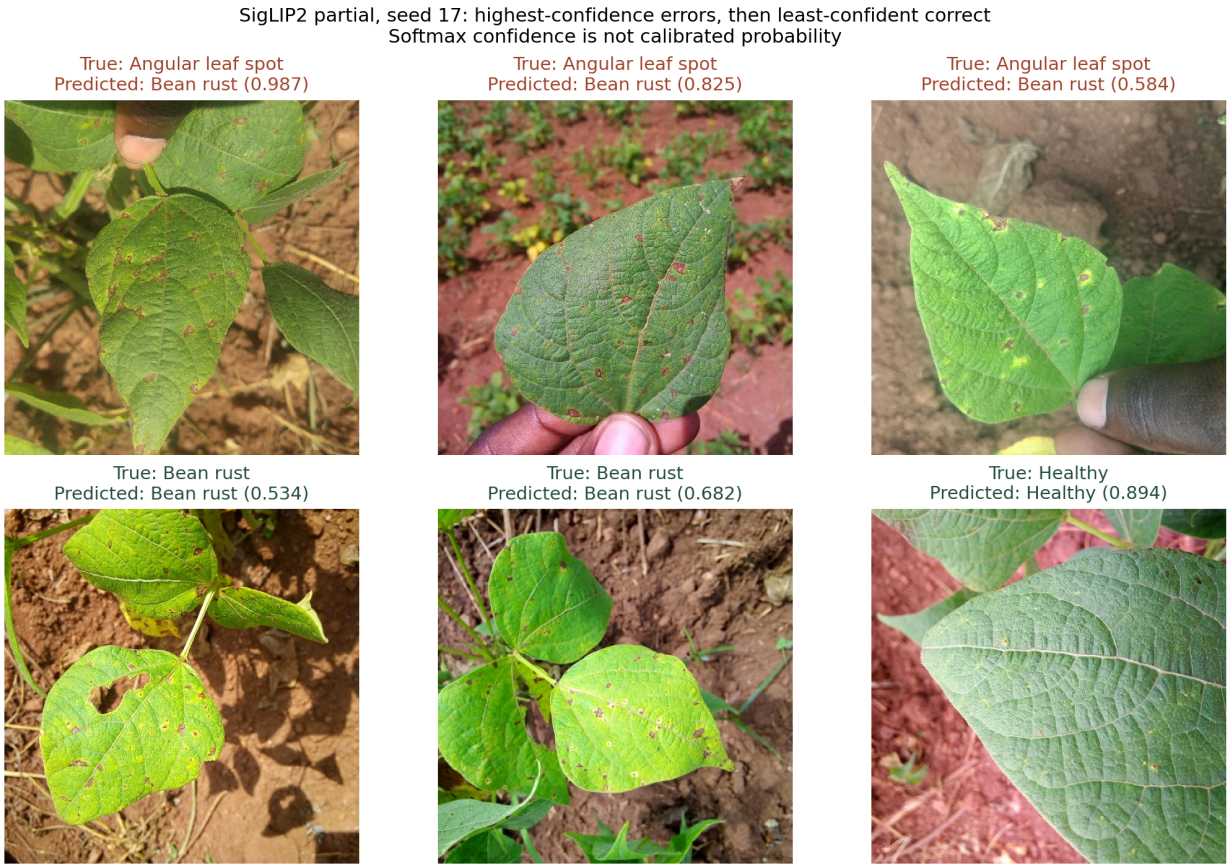


Figure 3. SigLIP 2 partial predictions: highest-confidence errors first, then lowest-confidence correct cases to fill six examples. Scores are uncalibrated.

7. Ethical and societal considerations

The potential benefit is accessible assistance with recognising visible symptoms where agronomic advice is difficult to obtain. Errors have unequal consequences: missed disease could delay expert assessment; false alarms could cause expense or inappropriate intervention. The reported harm-related counts are small-sample observations, not measured deployment outcomes. Zero observed disease-as-healthy errors does not establish zero risk. Any future system should assist agricultural experts and avoid automatically prescribing treatment.

The dataset represents one crop and two disease categories from a restricted collection. Other diseases, pests and nutrient problems would still be forced into one of three labels. Softmax confidence cannot establish correctness, especially outside the training distribution. Before deployment, prospective evaluation should cover independently collected farms, seasons, devices and disease presentations. A calibrated rejection or referral policy would need separate validation, and users should be able to obtain expert review rather than receiving an unsupported definitive diagnosis.

Balanced classes do not establish fairness across farmers. Farm, location and relevant demographic metadata are insufficient for subgroup evaluation here. The models' large pretraining collections may also encode unequal geographic or socioeconomic representation. SigLIP 2's reported fairness improvements concern separate benchmarks [1]; they do not certify fairness for this agricultural application. Testing performance across local collection conditions and involving agricultural practitioners would be necessary.

Phone quality, connectivity and access to advice could influence who benefits, even if model accuracy were similar.

Makerere AI Lab, expert annotators and model authors require attribution. Dataset MIT and model Apache-2.0 notices accompany the project; public availability does not itself demonstrate consent for unrelated future collection or deployment. New field data should minimise unnecessary personal or farm-identifying information. Reusing pretrained weights and restricting adaptation reduces the work performed in this experiment, but local runtime excludes the models' substantial pretraining costs and does not justify invented carbon savings. Transparent limitations and disclosure of AI assistance support accountable reporting rather than an exaggerated claim of readiness.

8. Conclusion

SigLIP 2 was accessible and feasible, and supplied stronger frozen features; the partial-model means were effectively tied. For this exploratory resource-limited setting, partial DINOv2 offers comparable accuracy with a smaller model and memory requirement. Testing recent models remains valuable without assuming improvement. Independent farm-level validation and validation-only tuning are the next priorities.

References

[1] Tschannen, M., et al. (2025). SigLIP 2: Multilingual Vision-Language Encoders with Improved Semantic Understanding, Localization, and Dense Features. Paper.

[2] Oquab, M., et al. (2023). DINOv2: Learning Robust Visual Features without Supervision. Paper.

[3] Makerere AI Lab (2020). Bean disease dataset. Repository.

[4] Makerere AI Lab (n.d.). Beans dataset card. Accessed 24 October 2026. Dataset.

[5] Google (2025). SigLIP 2 B/16-224 model and processor. Model card. Processor.

[6] Hugging Face (2025). Transformers 4.57.1 SigLIP implementation. Source.

[7] Meta AI (n.d.). DINOv2-Small configuration and processor. Configuration. Processor.